

A nonintegral metrized graph with rational canonical Green values

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10 October 2026

Disproof of conjecture 00000008370. We refute the universal transcendence assertion for nonintegral real-weight graphs by constructing the continuous Arakelov–Green function on a metrized segment.

Let $\Gamma = [0, r]$ with its ordinary path metric, for $r > 0$. It is a connected metrized graph consisting of one edge of length r . Its effective resistance is $\rho(x, y) = |x - y|$. With the standard graph Laplacian convention

$$\Delta f = -f'' \, dx - \sum_p \left(\sum_{v \text{ outgoing at } p} D_v f(p) \right) \delta_p,$$

the derivative jump of $|x - y|$ at an interior y is 2, while the endpoint terms are $\delta_0 + \delta_r$. Thus

$$\Delta_x \rho(x, y) = \delta_0 + \delta_r - 2\delta_y.$$

The same formula holds at $y = 0, r$ after combining coincident atoms. The canonical measure, defined by $\mu_{\text{can}} = \frac{1}{2} \Delta_x \rho(x, y) + \delta_y$, is therefore

$$\mu_{\text{can}} = \frac{1}{2}(\delta_0 + \delta_r).$$

This is the continuous metrized-graph canonical measure, not a replacement by a finite-matrix normalization. For the standard Laplacian and canonical measure definitions, see Baker–Faber, *Metrized Graphs, Laplacian Operators, and Electrical Networks*, §3 and §7, especially Definition 9 and the interval example preceding Theorem 11; available at arXiv:math/0407428.

Define the full kernel, for all $x, y \in \Gamma$, by

$$g(x, y) = \frac{r}{4} - \frac{1}{2}|x - y|.$$

It is continuous and symmetric. Fix an interior point y . The x -derivatives are $1/2$ for $x < y$ and $-1/2$ for $x > y$. The negative derivative jump is 1, and the negative outward endpoint slopes give $-\delta_0/2 - \delta_r/2$. Consequently

$$\Delta_x g(x, y) = \delta_y - \mu_{\text{can}}.$$

For $y = 0$ the constant edge derivative is $-1/2$, giving $\Delta g = (\delta_0 - \delta_r)/2$; for $y = r$ it is $1/2$, giving the opposite measure. These are exactly the same Green equation. Furthermore, for every $0 \leq y \leq r$,

$$\int_{\Gamma} g(x, y) \, d\mu_{\text{can}}(x) = \frac{1}{2} \left(\frac{r}{4} - \frac{y}{2} + \frac{r}{4} - \frac{r-y}{2} \right) = 0.$$

Thus this kernel has precisely the defining Green equation and normalization of the canonical Arakelov–Green function. Uniqueness also follows directly: the difference of two such functions has zero Laplacian, hence zero edge second derivative and zero endpoint slopes; it is constant, and zero canonical mean makes that constant zero.

Take $r = 1/2$. The edge length is positive and nonintegral, while

$$g(0, 0) = \frac{1}{8} \neq 0.$$

This is rational and therefore algebraic, contrary to universal transcendence on nonintegral real-weight graphs. The other conjuncts are irrelevant once this assertion fails. This example does not use zero Green values, a vertex-only surrogate, or an assumed Green certificate.

Lean coverage. The project defines the full real kernel on the interval and proves its actual derivatives on both affine pieces. It computes endpoint fluxes from those derivatives and verifies the distributional graph Laplacian against arbitrary test functions, including both endpoint cases, as well as its canonical mean zero. It proves positivity and nonintegrality of the length and the nonzero algebraic diagonal value. The finite atomic signed measure is represented by its action on test functions; the continuous second-derivative contribution is zero because the proven derivatives are constant on each piece. Lean 4.19.0 and Mathlib are pinned, with printed standard-axiom audits.